XCPC Programming Contest Problem Set

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December 4, 2025

Problem List

A	Hello, World!
B	A + B
C	Palindrome Check
D	Array Sorting
E	LaTeX Example (Demo)
F	Markdown Example (Demo)

This problem set should contain 6 problems on 6 numbered pages.
Please inform a runner immediately if something is missing from your problem set.

Problem A. Hello, World!

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Input file:	`standard input`
Output file:	`standard output`
Time Limit:	1 second
Memory Limit:	512 megabytes

Given a string S, output "Hello, S!"

Input

Input a string S.

Output

Output the answer.

Examples

standard input	standard output
`World`	`Hello, World!`
`Typst`	`Hello, Typst!`

Note

The length of the string does not exceed 100.

Problem B. A + B

Calculate the value of $A + B$.

Input

Input two integers A and B.

Output

Output the value of $A + B$.

Examples

standard input	standard output
1 2	3
10 20	30

Note

$-10^9 \leq A, B \leq 10^9$.

Problem C. Palindrome Check

Given a string, determine whether it is a palindrome (a string that reads the same forwards and backwards).

Input

Input a string S.

Output

Output `Yes` if it is a palindrome, otherwise output `No`.

Examples

standard input	standard output
racecar	Yes
hello	No
a	Yes

Note

The length of the string does not exceed 10^5.

Problem D. Array Sorting

Given an array, sort it and output the result.

Input

The first line contains an integer n ($1 \leq n \leq 1000$), representing the length of the array.
The second line contains n integers, representing the array elements.

Output

Output the sorted array, with elements separated by spaces.

Examples

standard input	standard output
5 3 1 4 2 5	1 2 3 4 5
3 5 5 5	5 5 5

Note

Array elements range from -10^6 to 10^6.

Problem E. LaTeX Example (Demo)

Calculate the factorial of n.
Definition: The factorial of a non-negative integer n is the product of all positive integers less than or equal to n.

Input

Input an integer n ($0 \leq n \leq 20$).

Output

Output the value of $n!$.

Examples

standard input	standard output
3	6
5	120

Note

Note that $20! = 2,432,902,008,176,640,000$.

Problem F. Markdown Example (Demo)

For a given integer n, output the squares of all integers from 1 to n.

Example:

- When $n = 3$, output `1 4 9`
- When $n = 5$, output `1 4 9 16 25`

Formula

The square of each integer can be calculated using the following formula:

$$x^2 = x \times x$$

where x is an integer from 1 to n.

Input

The first line contains an integer n ($1 \leq n \leq 1000$).

Output

Output n numbers, where the i-th number is i^2, separated by spaces.

Examples

standard input	standard output
5	1 4 9 16 25
3	1 4 9

Note

Data Range

- $1 \leq n \leq 1000$
- Results are within 32-bit integer range

Hint

You can use a loop or directly calculate using the formula.